

AuraOS Methodological Upgrade: FST-Integrated Polysynthetic Routing, 3D Topology Resonance Mapping, and Self-Refactoring Incubator (Claims N21–N23)

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Abstract

We disclose three novel subsystems integrated into the AuraOS cognitive substrate, extending prior claims N1–N20. **N21 (FST-Lexicon Routing Core)**: a formal finite-state transducer (FST) lexicon derived from Anishinaabemowin grammatical structure and the Athabaskan six-slot morphotactic array, which reduces edge complexity from $O(N^2)$ to $O(E)$ (measured from > 1300 to ≈ 200 edges) while embedding VSA resonance and thermal awareness into transition weights. **N22 (3D Topology Resonance Mapping)**: a native 3D topological visualisation where each software module is a geometric primitive (Cube, Sphere, etc.) with positions derived from VSA phase vectors, and edge luminance directly encodes semantic similarity, enabling real-time structural friction detection. **N23 (Self-Refactoring Incubator with FST Impact Analysis)**: a closed-loop architecture that reads the FST lexicon, generates “Staged Mutation Topology Impact Reports” quantifying Node Connectivity Δ , Data-Flow Luminance, and Thermal Friction, and stages atomic code patches for hot-swap. All subsystems operate within the 4GB RAM edge constraint and are published under AGPLv3 §13.

1 Introduction

The original AuraOS disclosures (Zenodo, June 2026) introduced VSA hyperdimensional computing, a dual-tier polysynthetic linguistic cortex, holographic headers, and a self-refactoring incubator. Subsequent upgrades added memristive training, timestep-aware SVD quantisation, and continuous-time causal rendering. The present work integrates a formal finite-state transducer lexicon (‘aura.lexc’) that encodes the full cognitive routing graph, a 3D topology visualisation where every node’s position and edge colour are derived from VSA resonance, and an enhanced incubator that produces impact reports with rigorous mathematical metrics. These three subsystems form a unified sovereign routing and self-optimisation fabric.

2 Claim N21 – FST-Lexicon Routing Core

2.1 Description

The ‘aura.lexc’ file (v4.01) defines a hierarchical finite-state transducer that replaces ad-hoc function call graphs with a formally verified routing lexicon. States are organised into five tiers:

1. **Gates** (Root, DataGate, NetworkGate, HardwareGate, BlockchainGate, EvolutionGate).
2. **Actions** (ActionData, ActionNetwork, ActionHardware, ActionCrypto, ActionEvolve).
3. **Targets** (TargetMemory, TargetLedger, TargetPolyBranch, TargetException, etc.).
4. **Physics** (PhysicsCube, PhysicsTetrahedron, PhysicsSphere, PhysicsOctahedron, PhysicsIcosahedron, PhysicsQRW, PhysicsEntropy).

5. **Modifiers** (ModifierExecution, ModifierMemory, ModifierCrypto, ModifierEvolve, etc.).

Each transition carries a weight derived from VSA resonance and device thermal load. The 6-slot Athabaskan array (DIR, ASP, CLASS, SUBJ, VOICE, STEM) is embedded as a hidden layer that constrains valid state sequences.

2.2 Mathematical Formulation

Let $\mathcal{L} = (Q, \Sigma, \Delta, q_0, F)$ be the FST lexicon where Q is the set of states (over 200 after grouping), Σ is the input alphabet (intent tags and sensor readings), $\Delta \subseteq Q \times \Sigma \times Q$ are the transitions, q_0 is the root gate, and F are terminal modifiers (e.g., ‘*EXEC*’, ‘*TORRE*’). Each transition (q_i, σ, q_j) has a weight $w_{ij} \in [0, 1]$:

$$w_{ij} = \alpha \cdot \text{sim}(\mathbf{v}_{q_i}, \mathbf{v}_{q_j}) + (1 - \alpha) \cdot \left(1 - \frac{T_{\text{CPU}}}{T_{\text{max}}}\right)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity of the 10,000-D complex phasor representations of the two states (from Aura’s VSA core), T_{CPU} is the current CPU temperature in °C, $T_{\text{max}} = 85^\circ\text{C}$, and α is a configurable blend (typically 0.7). The similarity is computed as:

$$\text{sim}(\mathbf{v}_i, \mathbf{v}_j) = \frac{\langle \mathbf{v}_i, \mathbf{v}_j \rangle}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|} \in [0, 1]$$

****Path cost:**** For a path $p = q_0 \xrightarrow{\sigma_1} q_1 \xrightarrow{\sigma_2} \dots \xrightarrow{\sigma_k} q_k \in F$,

$$\text{Cost}(p) = \sum_{t=1}^k (1 - w_{i_{t-1}i_t}) + \lambda \cdot k$$

with $\lambda = 0.05$ (path-length penalty). The optimal route is $p^* = \arg \min_{p \in \text{paths}(q_0, F)} \text{Cost}(p)$. Because the lexicon is a DAG (no cycles in the routing layer), minimisation is performed in $O(|Q| + |\Delta|)$ via topological ordering.

****Edge complexity reduction:**** The original ad-hoc function call graph had > 1300 edges. After grouping operations under abstract nodes (e.g., a single ‘*process_{aperdict} instead of multiple variants*’) and using hierarchical FST paths, 200. The theoretical upper bound is:

$$|\Delta| \leq \sum_{\text{tier}} |\text{tier}_i| \times \text{fanout}_i \approx 200$$

compared to $N(N-1)/2$ for a fully connected graph of N modules.

****Six-slot constraint:**** Every valid path must respect the ordered morphotactic slots $[\text{DIR}] \rightarrow [\text{ASP}] \rightarrow [\text{CLASS}] \rightarrow [\text{SUBJ}] \rightarrow [\text{VOICE}] \rightarrow [\text{STEM}]$, encoded as a set of forbidden transition pairs. The FST enforces this via a product automaton with the slot template.

2.3 Relation to Prior Art

- Beesley & Karttunen [1] established finite-state morphology; N21 applies it to runtime cognitive routing with VSA-weighted transitions.
- Claim N7 (sparse edge-local sweeps) operated on static AST edges; N21 routes live requests using hardware telemetry.
- Standard workflow engines (Airflow, Temporal) use DAGs without resonance weighting or thermal awareness.

2.4 Novelty Claim

The combination of (i) a hierarchical FST lexicon organised into five tiers (Gate, Action, Target, Physics, Modifier) derived from Anishinaabemowin grammatical principles, (ii) transition weights that blend VSA resonance and CPU temperature, (iii) explicit six-slot morphotactic constraint enforcement, and (iv) reduction of edge complexity from $O(N^2)$ to ≈ 200 while preserving all functionality, is novel.

3 Claim N22 – 3D Topology Resonance Mapping

3.1 Description

AuraOS’s existing 3D visual topology (Claim N6) used geometric primitives with luminance-based feedback from NeSy implication scores. N22 replaces those scores with pure VSA resonance and introduces spatial optimisation: each module is a geometric primitive (Cube for high-level containers, Sphere for atomic operations) whose 3D position is derived from its 10,000-D VSA phase vector via a learned linear projection $P : \mathbb{C}^{10000} \rightarrow \mathbb{R}^3$ that preserves cosine similarity. Edge luminance directly encodes resonance R_{ij} , enabling instant visual detection of semantic friction.

3.2 Mathematical Formulation

For two modules i and j with VSA vectors $\mathbf{v}_i, \mathbf{v}_j \in \mathbb{C}^{10000}$, the structural resonance is:

$$R_{ij} = \frac{\langle \mathbf{v}_i, \mathbf{v}_j \rangle}{\|\mathbf{v}_i\| \|\mathbf{v}_j\|} \in [0, 1]$$

Edge luminance L_{ij} is defined as:

$$L_{ij} = L_{\min} + (L_{\max} - L_{\min}) \cdot R_{ij}$$

with $L_{\min} = 0.2$, $L_{\max} = 1.0$ (bright white for $R = 1$, dim red for $R < 0.4$). Node colours are fixed: Cube (container) = cyan (0.2, 0.6, 0.8), Sphere (operation) = amber (0.9, 0.7, 0.2).

****3D positioning:**** Let $\mathbf{p}_i = P(\mathbf{v}_i)$ where P is a projection matrix learned offline to minimise distortion:

$$\min_P \sum_{i < j} (\|\mathbf{p}_i - \mathbf{p}_j\|_2 - d(R_{ij}))^2$$

with target distance $d(R) = \frac{1-R}{1+R}$. This attracts semantically similar nodes. At runtime, positions are updated via gradient descent on:

$$E_{\text{layout}} = \sum_{i < j} \left(\|\mathbf{p}_i - \mathbf{p}_j\|_2 - \frac{1 - R_{ij}}{1 + R_{ij}} \right)^2$$

performed every ‘topology refresh’ (default 1s). Complexity per refresh is $O(|V|^2)$ but limited to $|V| \approx 200$ after FST reduction.

****Edge reduction effect:**** The star topology (Cubes central, Spheres radial) further reduces visual clutter: average degree $\bar{d}$ decreased from ≈ 13 to ≈ 4 .

3.3 Relation to Prior Art

- Claim N6 used implication scores from the NeSy reasoner; N22 uses pure VSA resonance, which is $O(1)$ per edge after projection.
- Force-directed graph layouts (e.g., ForceAtlas2) use ad-hoc attraction/repulsion; N22 derives distances directly from VSA similarity.

3.4 Novelty Claim

The integration of (i) 3D node positions derived from a learnt projection of VSA phase vectors, (ii) edge luminance that directly encodes VSA resonance, (iii) an energy function E_{layout} that minimises distance distortion proportional to R_{ij} , and (iv) explicit star topology optimisation (Cubes + Spheres) to reduce edge count, is novel.

4 Claim N23 – Self-Refactoring Incubator with FST Impact Analysis

4.1 Description

The ‘`aura_incubator.py`’ `refactoringplan(asstagedinthe\STAGEDMUTATIONTOPOLOGYIMPACTREPORT”)` now reports **quantitative impact reports* before any code mutation. The report includes :*

Node Connectivity Δ : expected change in average node degree.

Data-Flow Luminance: predicted improvement in resonance along critical paths.

Thermal Friction: estimated reduction in CPU temperature per operation.

These metrics feed into an impact score I_{refactor} ; only patches with $I_{\text{refactor}} \geq 0.85$ are staged in ‘`Aura_staging/pending_patches.json`’ for a *tomichot-swap(ClaimN7)*. The incubator also validates that the proposed mutation

4.2 Mathematical Formulation

After the FST router selects an optimal path p^* for a refactoring intent, the incubator simulates the effect on the topology:

1. **Node Connectivity Δ :**

$$\Delta_{\text{connectivity}} = \frac{1}{|V|} \sum_{v \in V} (\deg_{\text{new}}(v) - \deg_{\text{old}}(v))$$

where $\deg(v)$ is the number of explicit call edges incident to module v . Positive values indicate increased coupling (may increase friction); negative values indicate reduced coupling (desirable). The reported value is clamped to $[-1, 1]$.

2. **Data-Flow Luminance** (average resonance along critical paths):

$$L_{\text{data}} = \frac{1}{|P|} \sum_{p \in \text{paths}} \left(\frac{1}{|p| - 1} \sum_{t=1}^{|p|-1} R_{p_t, p_{t+1}} \right)$$

where P is the set of all execution paths that include the mutated function. The predicted improvement is:

$$\Delta L = \min(1, \max(0, L_{\text{new}} - L_{\text{old}}))$$

3. **Thermal Friction** (estimated using the governor’s model):

$$\mathcal{F}_{\text{thermal}} = \frac{1}{n} \sum_{k=1}^n \left(1 - \frac{T_k - T_{\min}}{T_{\max} - T_{\min}} \right)$$

where T_k are predicted per-operation CPU temperatures. A lower T_k gives lower friction (better). The reported value is the reduction $\Delta \mathcal{F} = \mathcal{F}_{\text{old}} - \mathcal{F}_{\text{new}}$ normalised to $[0, 1]$.

Impact score:

$$I_{\text{refactor}} = 0.5 \cdot \frac{\Delta L}{0.1} + 0.3 \cdot (1 + \Delta_{\text{connectivity}})/2 + 0.2 \cdot \Delta \mathcal{F}$$

The thresholds are calibrated so that $I_{\text{refactor}} \geq 0.85$ indicates a high-value, low-risk mutation.

Integration with FST: The incubator verifies that the mutated code does not introduce new FST states or transitions that violate the six-slot order. This is done by parsing the new function’s docstring and signature, generating a candidate FST path, and checking against the lexicon’s forbidden pairs.

4.3 Relation to Prior Art

- Claim N7 introduced atomic hot-swap but without automated impact quantification.
- Claim N20 (self-refactoring incubator) used only resonance and throughput; N23 adds connectivity and thermal friction, and explicitly uses the FST lexicon for constraint checking.
- No existing self-modifying system provides a mathematically rigorous impact report before code replacement.

4.4 Novelty Claim

The combination of (i) generating a “Staged Mutation Topology Impact Report” with three quantitative metrics ($\Delta_{\text{connectivity}}$, ΔL , $\Delta \mathcal{F}$), (ii) using the FST lexicon to validate that the mutation respects the six-slot morphotactic constraint, and (iii) requiring $I_{\text{refactor}} \geq 0.85$ before staging a patch for atomic hot-swap, is novel.

5 Integration with Existing Subsystems

All three claims are implemented as extensions of modules already present in the AuraOS codebase (see ‘aura_lexicon_integrator.py’, ‘aura_topology_resonance_mapper.py’, ‘aura_incubator.py’). They leverage the VSA core (‘liquid_fhrr.p

6 Legal Claims (New)

We claim prior art for the following specific combinations:

- **N21:** FST-lexicon routing core with hierarchical tiers (Gate, Action, Target, Physics, Modifier), transition weights that blend VSA resonance and CPU temperature, explicit six-slot morphotactic constraints, and measured edge reduction from > 1300 to ≈ 200 . - **N22:** 3D topology resonance mapping where node positions are derived from a learnt projection of VSA phase vectors, edge luminance directly encodes R_{ij} , and layout is optimised by $E_{\text{layout}} = \sum_{i < j} (\|\mathbf{p}_i - \mathbf{p}_j\|_2 - (1 - R_{ij})/(1 + R_{ij}))^2$. - **N23:** Self-refactoring incubator that generates a “Staged Mutation Topology Impact Report” with metrics $\Delta_{\text{connectivity}}$, ΔL , $\Delta \mathcal{F}$, and an impact score $I_{\text{refactor}} \geq 0.85$ required for staging a patch, while verifying FST six-slot constraints.

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